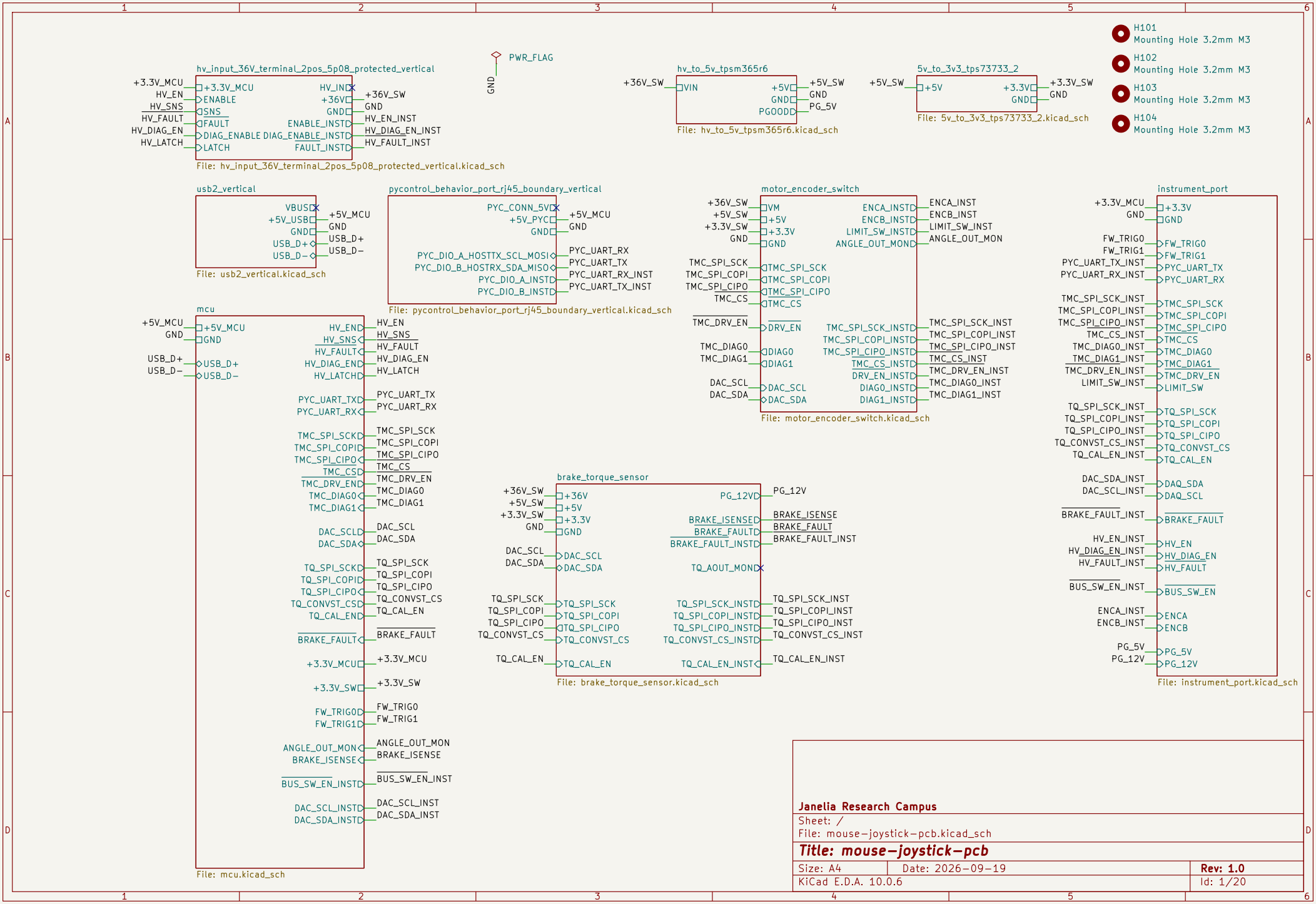
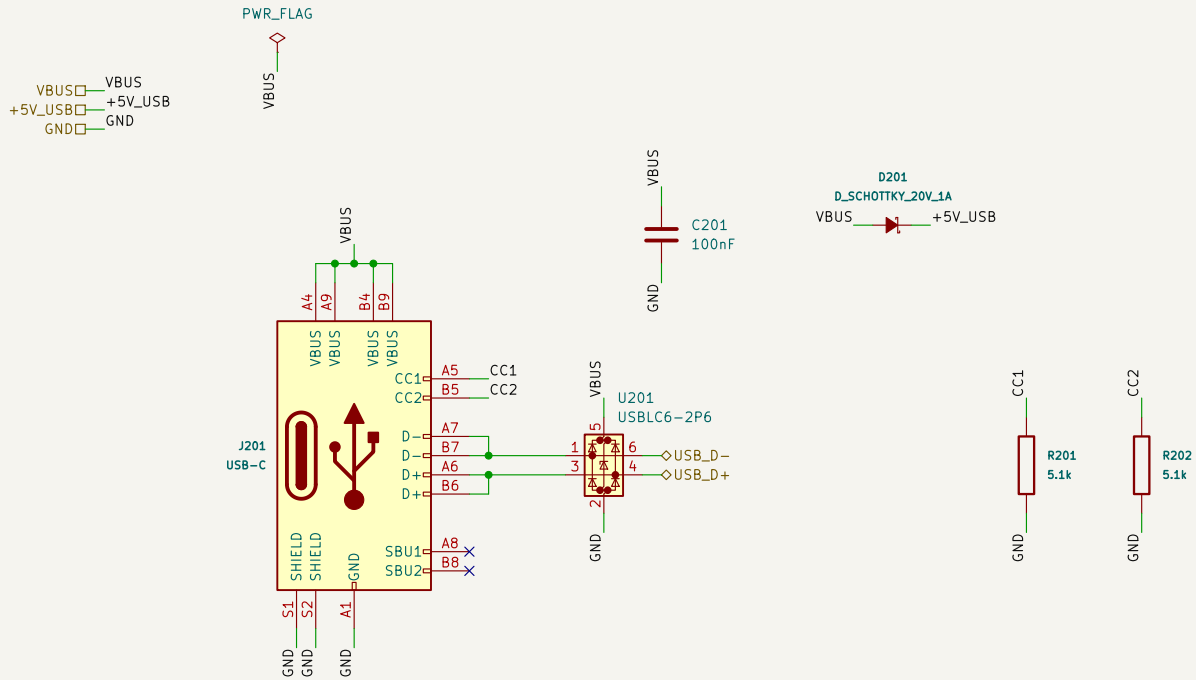

hv_input_36V_terminal_2pos_5p08_protected_vertical
+3.3V_MCU
HV_EN
HV_SNS
HV_FAULT
HV_DIAG_EN
HV_LATCH
+3.3V_MCU
ENABLE
SNS
FAULT
DIAG_ENABLE
LATCH
HV_IN
+36V
GND
ENABLE_INST
DIAG_ENABLE_INST
FAULT_INST
+36V_SW
GND
HV_EN_INST
HV_DIAG_EN_INST
HV_FAULT_INST
File: hv_input_36V_terminal_2pos_5p08_protected_vertical.kicad_sch
PWR_FLAG
GND
hv_to_5v_tpsm365r6
+36V_SW
VIN
+5V
GND
PGOOD
+5V_SW
GND
PG_5V
File: hv_to_5v_tpsm365r6.kicad_sch
5v_to_3v3_tps73733_2
+5V_SW
+5V
+3.3V
GND
+3.3V_SW
GND
File: 5v_to_3v3_tps73733_2.kicad_sch
H101
Mounting Hole 3.2mm M3
H102
Mounting Hole 3.2mm M3
H103
Mounting Hole 3.2mm M3
H104
Mounting Hole 3.2mm M3
usb2_vertical
VBUS
+5V_USB
GND
USB_D+
USB_D−
+5V_MCU
GND
USB_D+
USB_D−
File: usb2_vertical.kicad_sch
pycontrol_behavior_port_rj45_boundary_vertical
PYC_CONN_5V
+5V_PYC
GND
PYC_DIO_A_HOSTTX_SCL_MOSI
PYC_DIO_B_HOSTRX_SDA_MISO
PYC_DIO_A_INST
PYC_DIO_B_INST
+5V_MCU
GND
PYC_UART_RX
PYC_UART_TX
PYC_UART_RX_INST
PYC_UART_TX_INST
File: pycontrol_behavior_port_rj45_boundary_vertical.kicad_sch
motor_encoder_switch
+36V_SW
+5V_SW
+3.3V_SW
GND
TMC_SPI_SCK
TMC_SPI_COPI
TMC_SPI_CIPO
TMC_CS
TMC_DRV_EN
TMC_DIAG0
TMC_DIAG1
DAC_SCL
DAC_SDA
VM
+5V
+3.3V
GND
DRV_EN
DIAG0
DIAG1
DAC_SCL
DAC_SDA
ENCA_INST
ENCB_INST
LIMIT_SW_INST
ANGLE_OUT_MOND
TMC_SPI_SCK_INST
TMC_SPI_COPI_INST
TMC_SPI_CIPO_INST
TMC_CS_INST
DRV_EN_INST
DIAG0_INST
DIAG1_INST
ENCA_INST
ENCB_INST
LIMIT_SW_INST
ANGLE_OUT_MON
TMC_SPI_SCK_INST
TMC_SPI_COPI_INST
TMC_SPI_CIPO_INST
TMC_CS_INST
TMC_DRV_EN_INST
TMC_DIAG0_INST
TMC_DIAG1_INST
File: motor_encoder_switch.kicad_sch
instrument_port
+3.3V_MCU
GND
FW_TRIG0
FW_TRIG1
PYC_UART_TX_INST
PYC_UART_RX_INST
TMC_SPI_SCK_INST
TMC_SPI_COPI_INST
TMC_SPI_CIPO_INST
TMC_CS_INST
TMC_DIAG0_INST
TMC_DIAG1_INST
TMC_DRV_EN_INST
LIMIT_SW_INST
TQ_SPI_SCK_INST
TQ_SPI_COPI_INST
TQ_SPI_CIPO_INST
TQ_CONVST_CS_INST
TQ_CAL_EN_INST
DAC_SDA_INST
DAC_SCL_INST
BRAKE_FAULT_INST
HV_EN_INST
HV_DIAG_EN_INST
HV_FAULT_INST
BUS_SW_EN_INST
ENCA_INST
ENCB_INST
PG_5V
PG_12V
+3.3V
GND
FW_TRIG0
FW_TRIG1
PYC_UART_TX
PYC_UART_RX
TMC_SPI_SCK
TMC_SPI_COPI
TMC_SPI_CIPO
TMC_CS
TMC_DIAG0
TMC_DIAG1
TMC_DRV_EN
LIMIT_SW
TQ_SPI_SCK
TQ_SPI_COPI
TQ_SPI_CIPO
TQ_CONVST_CS
TQ_CAL_EN
DAQ_SDA
DAQ_SCL
BRAKE_FAULT
HV_EN
HV_DIAG_EN
HV_FAULT
BUS_SW_EN
ENCA
ENCB
PG_5V
PG_12V
File: instrument_port.kicad_sch
mcu
+5V_MCU
GND
USB_D+
USB_D−
+5V_MCU
GND
USB_D+
USB_D−
HV_EN
HV_SNS
HV_FAULT
HV_DIAG_EN
HV_LATCH
PYC_UART_TX
PYC_UART_RX
TMC_SPI_SCK
TMC_SPI_COPI
TMC_SPI_CIPO
TMC_CS
TMC_DRV_EN
TMC_DIAG0
TMC_DIAG1
DAC_SCL
DAC_SDA
TQ_SPI_SCK
TQ_SPI_COPI
TQ_SPI_CIPO
TQ_CONVST_CS
TQ_CAL_EN
BRAKE_FAULT
+3.3V_MCU
+3.3V_SW
FW_TRIG0
FW_TRIG1
ANGLE_OUT_MON
BRAKE_ISENSE
BUS_SW_EN_INST
DAC_SCL_INST
DAC_SDA_INST
File: mcu.kicad_sch
brake_torque_sensor
+36V_SW
+5V_SW
+3.3V_SW
GND
DAC_SCL
DAC_SDA
TQ_SPI_SCK
TQ_SPI_COPI
TQ_SPI_CIPO
TQ_CONVST_CS
TQ_CAL_EN
+36V
+5V
+3.3V
GND
DAC_SCL
DAC_SDA
TQ_SPI_SCK
TQ_SPI_COPI
TQ_SPI_CIPO
TQ_CONVST_CS
TQ_CAL_EN
PG_12V
BRAKE_ISENSED
BRAKE_FAULT
BRAKE_FAULT_INST
TQ_AOUT_MON
TQ_SPI_SCK_INST
TQ_SPI_COPI_INST
TQ_SPI_CIPO_INST
TQ_CONVST_CS_INST
TQ_CAL_EN_INST
PG_12V
BRAKE_ISENSE
BRAKE_FAULT
BRAKE_FAULT_INST
TQ_SPI_SCK_INST
TQ_SPI_COPI_INST
TQ_SPI_CIPO_INST
TQ_CONVST_CS_INST
TQ_CAL_EN_INST
File: brake_torque_sensor.kicad_sch
Janelia Research Campus
Sheet: /
File: mouse−joystick−pcb.kicad_sch
Title: mouse−joystick−pcb
Size: A4
Date: 2026−09−19
Rev: 1.0
KiCad E.D.A. 10.0.6
Id: 1/20



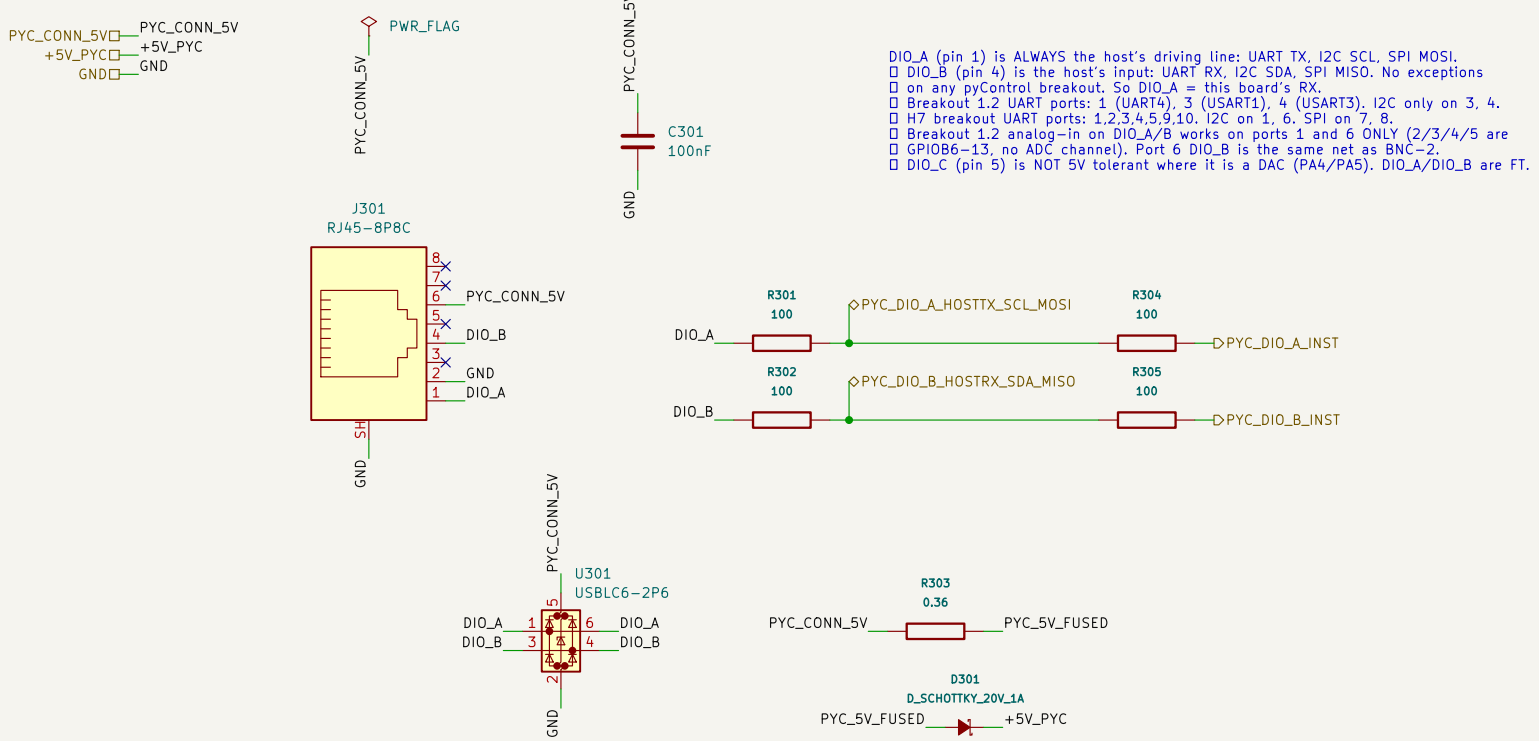
Copy or recreate old project circuits here, then save cleaned selections as design blocks.

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Sheet: /usb2_vertical/
File: usb2_vertical.kicad_sch

Title: mouse-joystick-pcb

Size: A4	Date: 2026-09-19	**Rev: 1.0**
KiCad E.D.A. 10.0.6		Id: 2/20

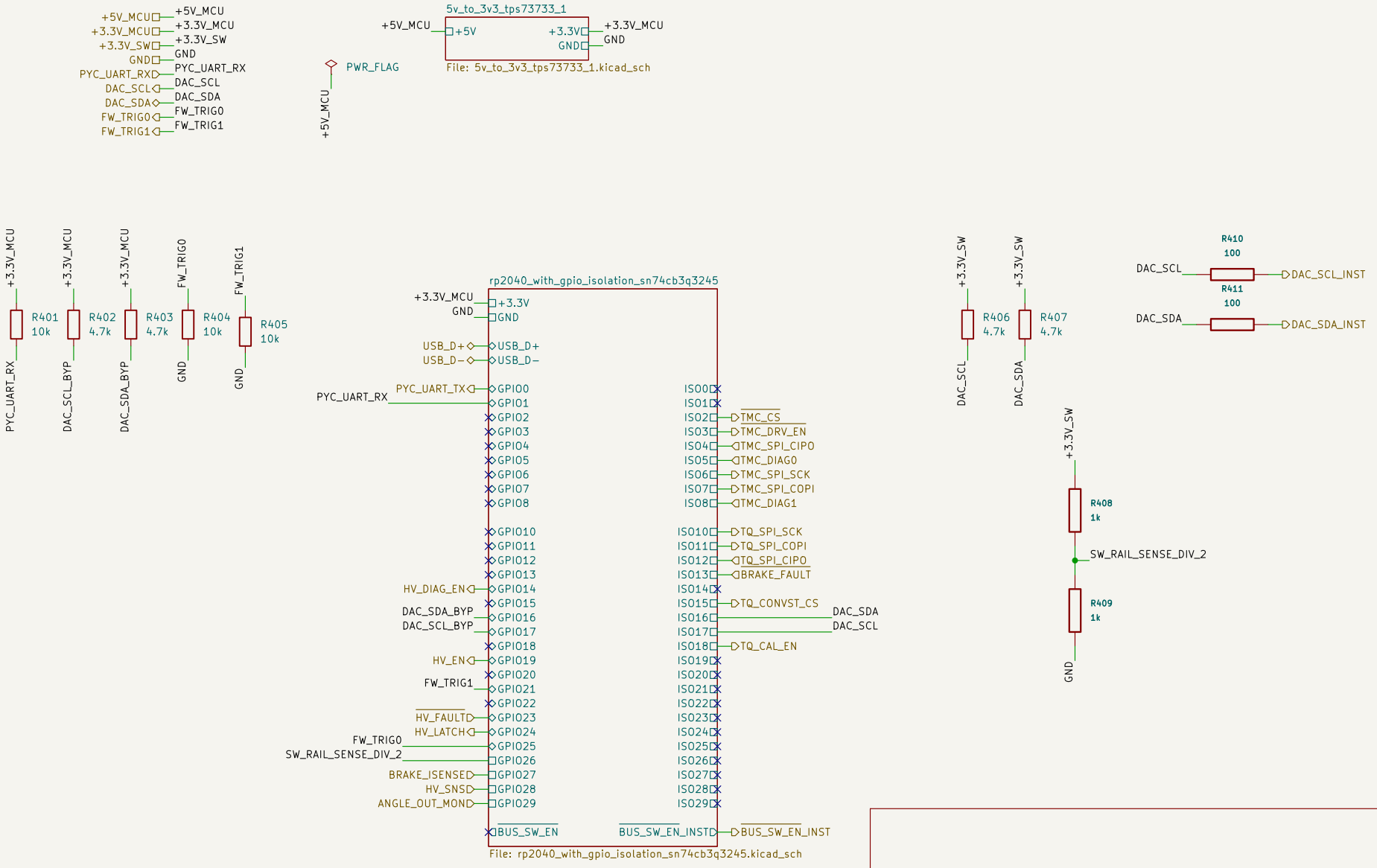


Copy or recreate old project circuits here, then save cleaned selections as design blocks.

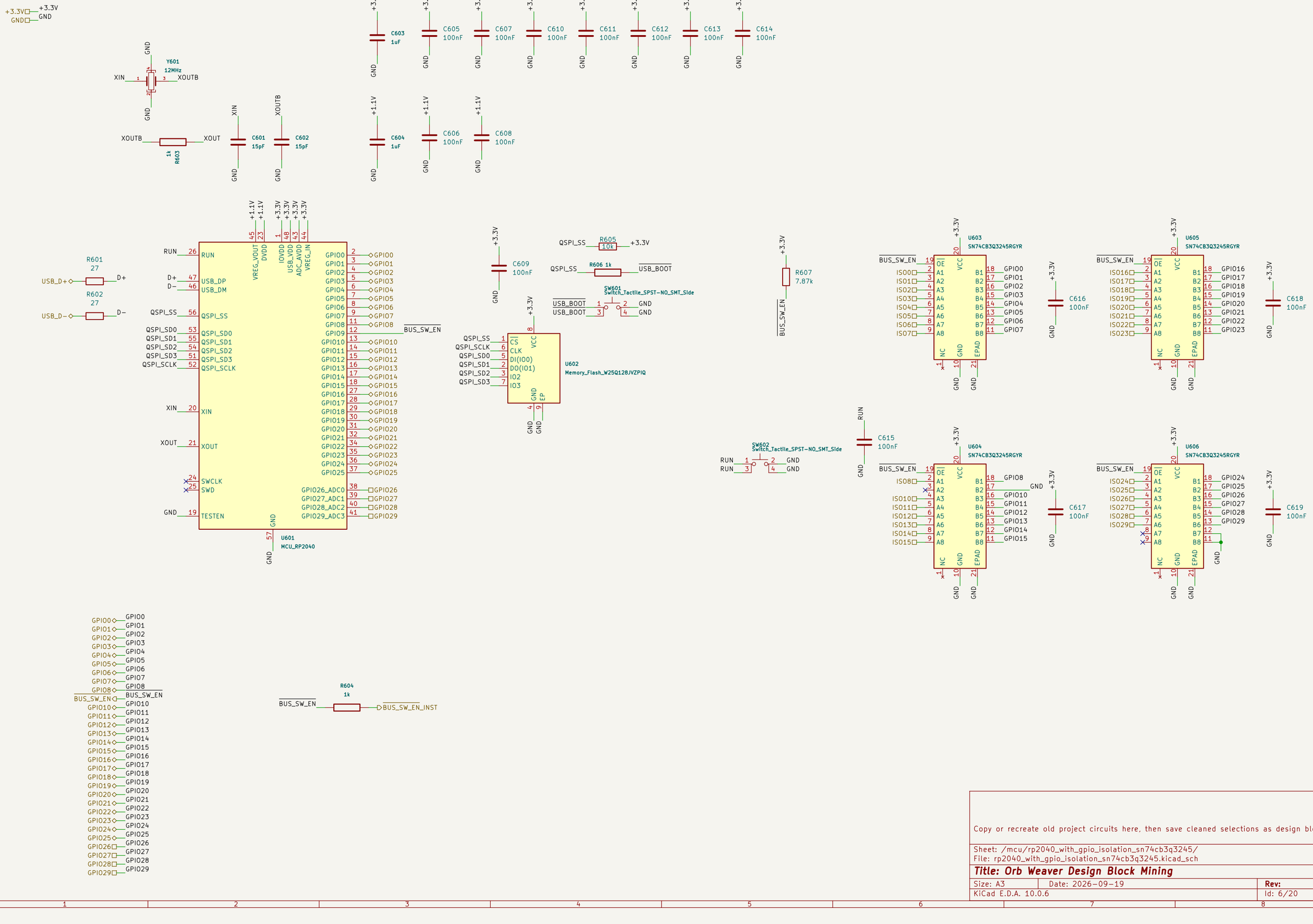
Sheet: /pycontrol_behavior_port_rj45_boundary_vertical/
File: pycontrol_behavior_port_rj45_boundary_vertical.kicad_sch

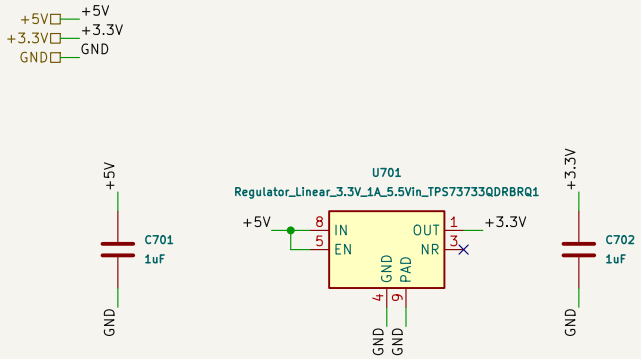
Title: Orb Weaver Design Block Mining

Size: A4	Date: 2026-09-19	**Rev:**
KiCad E.D.A. 10.0.6		Id: 3/20



Janelia Research Campus		
Sheet: /mcu/		
File: mcu.kicad_sch		
Title: mouse−joystick−pcb		
Size: A4	Date: 2026−09−19	**Rev: 1.0**
KiCad E.D.A. 10.0.6		Id: 4/20


+3.3V
GND
Y601
12MHz
XIN
XOUTB
R603
1k
C601
15pF
C602
15pF
C603
1uF
C604
1uF
C605
100nF
C606
100nF
C607
100nF
C608
100nF
C609
100nF
C610
100nF
C611
100nF
C612
100nF
C613
100nF
C614
100nF
R601
27
R602
27
USB_D+
USB_D−
D+
D−
R605
10k
R606 1k
QSPI_SS
USB_BOOT
SW601
Switch_Tactile_SPST-NO_SMT_Side
U601
MCU_RP2040
RUN
XIN
XOUT
SWCLK
SWD
TESTEN
QSPI_SS
QSPI_SD0
QSPI_SD1
QSPI_SD2
QSPI_SD3
QSPI_SCLK
BUS_SW_EN
U602
Memory_Flash_W25Q128JVZPIQ
R607
7.87k
U603
SN74CB3Q3245RGYR
U604
SN74CB3Q3245RGYR
U605
SN74CB3Q3245RGYR
U606
SN74CB3Q3245RGYR
C615
100nF
C616
100nF
C617
100nF
C618
100nF
C619
100nF
SW602
Switch_Tactile_SPST-NO_SMT_Side
R604
1k
BUS_SW_EN_INST
Copy or recreate old project circuits here, then save cleaned selections as design blocks.
Sheet: /mcu/rp2040_with_gpio_isolation_sn74cb3q3245/
File: rp2040_with_gpio_isolation_sn74cb3q3245.kicad_sch
Title: Orb Weaver Design Block Mining
Size: A3
Date: 2026-09-19
Rev:
KiCad E.D.A. 10.0.6
Id: 6/20



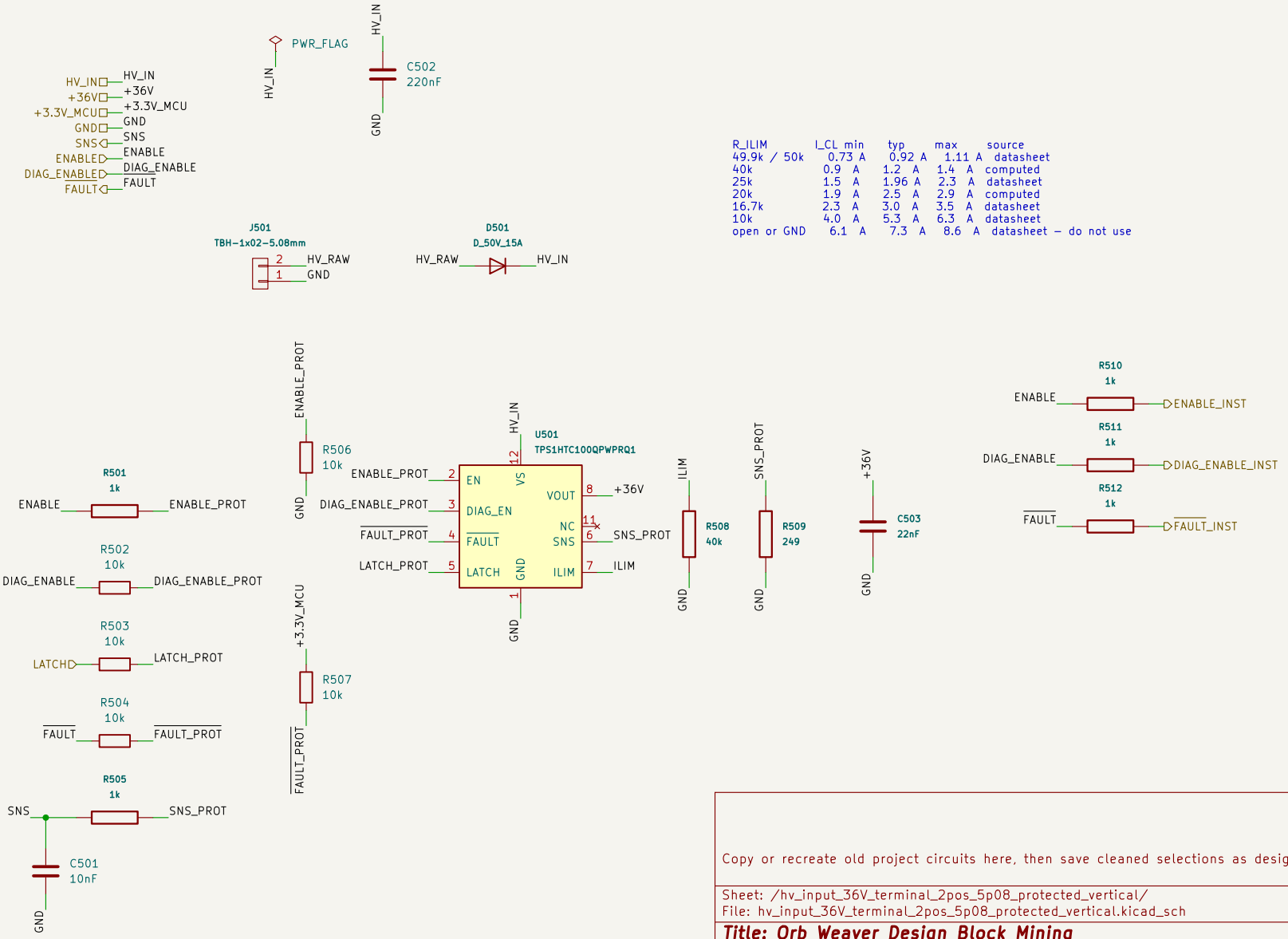
Copy or recreate old project circuits here, then save cleaned selections as design blocks.

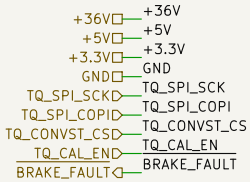
Janelia Research Campus

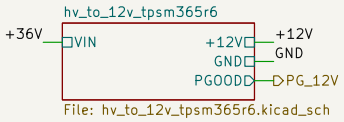
Sheet: /mcu/5v_to_3v3_tps73733_1/
File: 5v_to_3v3_tps73733_1.kicad_sch

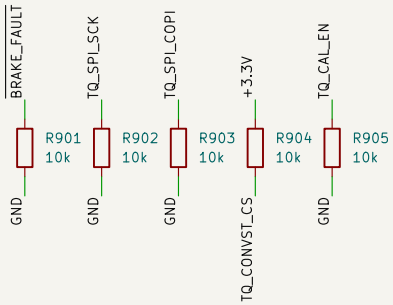
Title: mouse-joystick-pcb

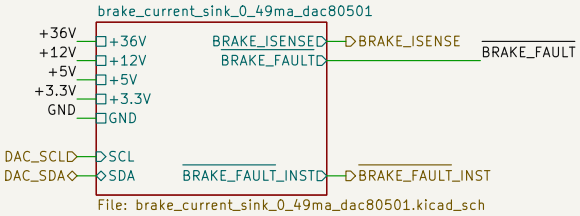
Size: A4	Date: 2026-09-19	**Rev: 1.0**
KiCad E.D.A. 10.0.6		Id: 7/20

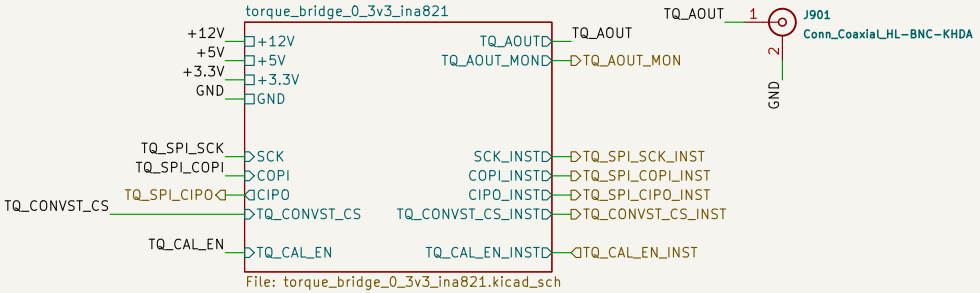

HV_IN
+36V
+3.3V_MCU
GND
SNS
ENABLE
DIAG_ENABLE
FAULT
PWR_FLAG
C502
220nF
R_ILIM I_CL min typ max source
49.9k / 50k 0.73 A 0.92 A 1.11 A datasheet
40k 0.9 A 1.2 A 1.4 A computed
25k 1.5 A 1.96 A 2.3 A datasheet
20k 1.9 A 2.5 A 2.9 A computed
16.7k 2.3 A 3.0 A 3.5 A datasheet
10k 4.0 A 5.3 A 6.3 A datasheet
open or GND 6.1 A 7.3 A 8.6 A datasheet — do not use
J501
TBH-1x02-5.08mm
HV_RAW
GND
D501
D_50V_15A
U501
TPS1HTC100QPWPRQ1
EN
DIAG_EN
FAULT
LATCH
VS
GND
VOUT
NC
SNS
ILIM
R501 1k
R502 10k
R503 10k
R504 10k
R505 1k
R506 10k
R507 10k
R508 40k
R509 249
R510 1k
R511 1k
R512 1k
C501 10nF
C503 22nF
ENABLE_PROT
DIAG_ENABLE_PROT
FAULT_PROT
LATCH_PROT
SNS_PROT
ENABLE_INST
DIAG_ENABLE_INST
FAULT_INST
Copy or recreate old project circuits here, then save cleaned selections as design blocks.
Sheet: /hv_input_36V_terminal_2pos_5p08_protected_vertical/
File: hv_input_36V_terminal_2pos_5p08_protected_vertical.kicad_sch
Title: Orb Weaver Design Block Mining
Size: A4
Date: 2026-09-19
Rev:
KiCad E.D.A. 10.0.6
Id: 5/20



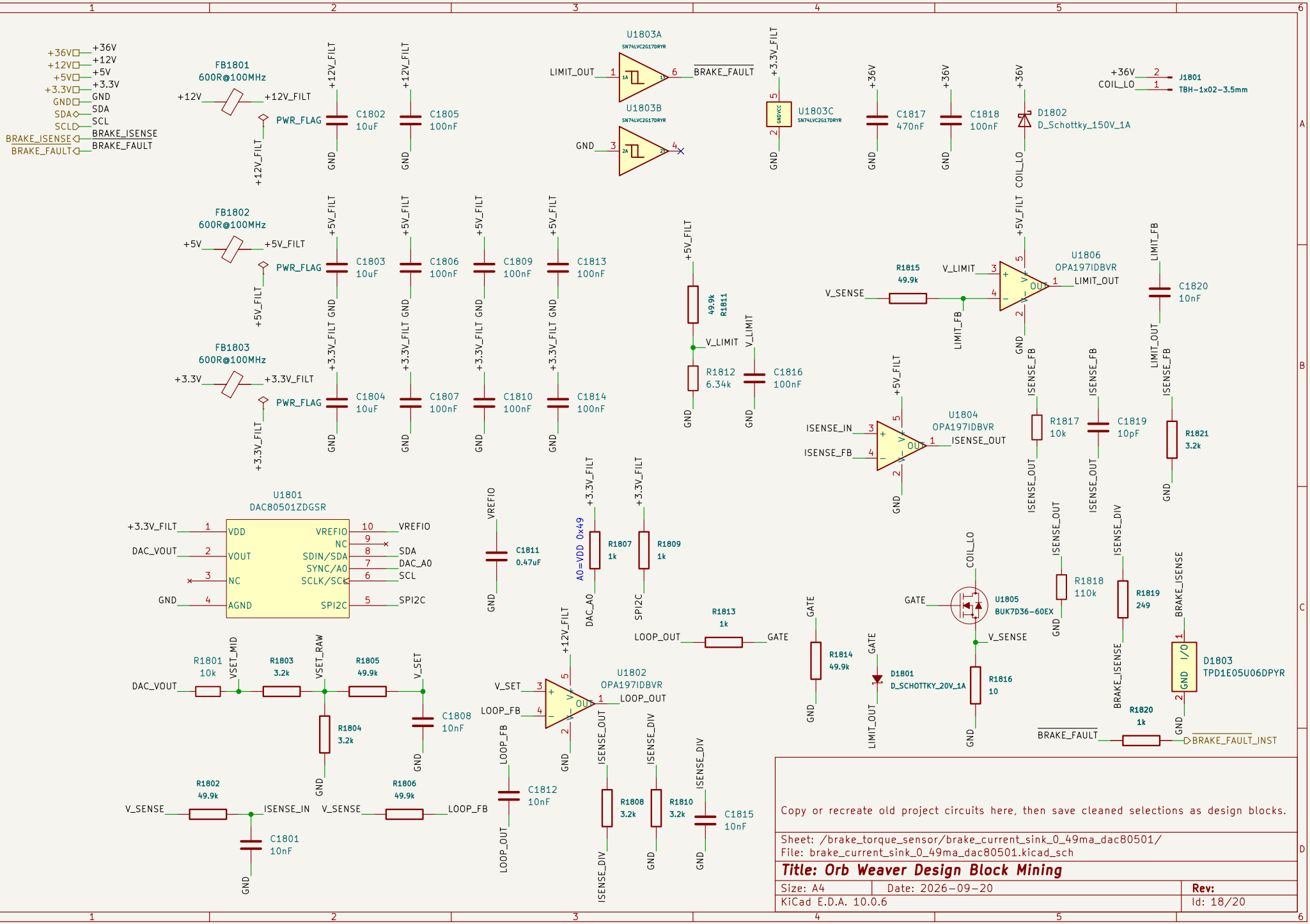


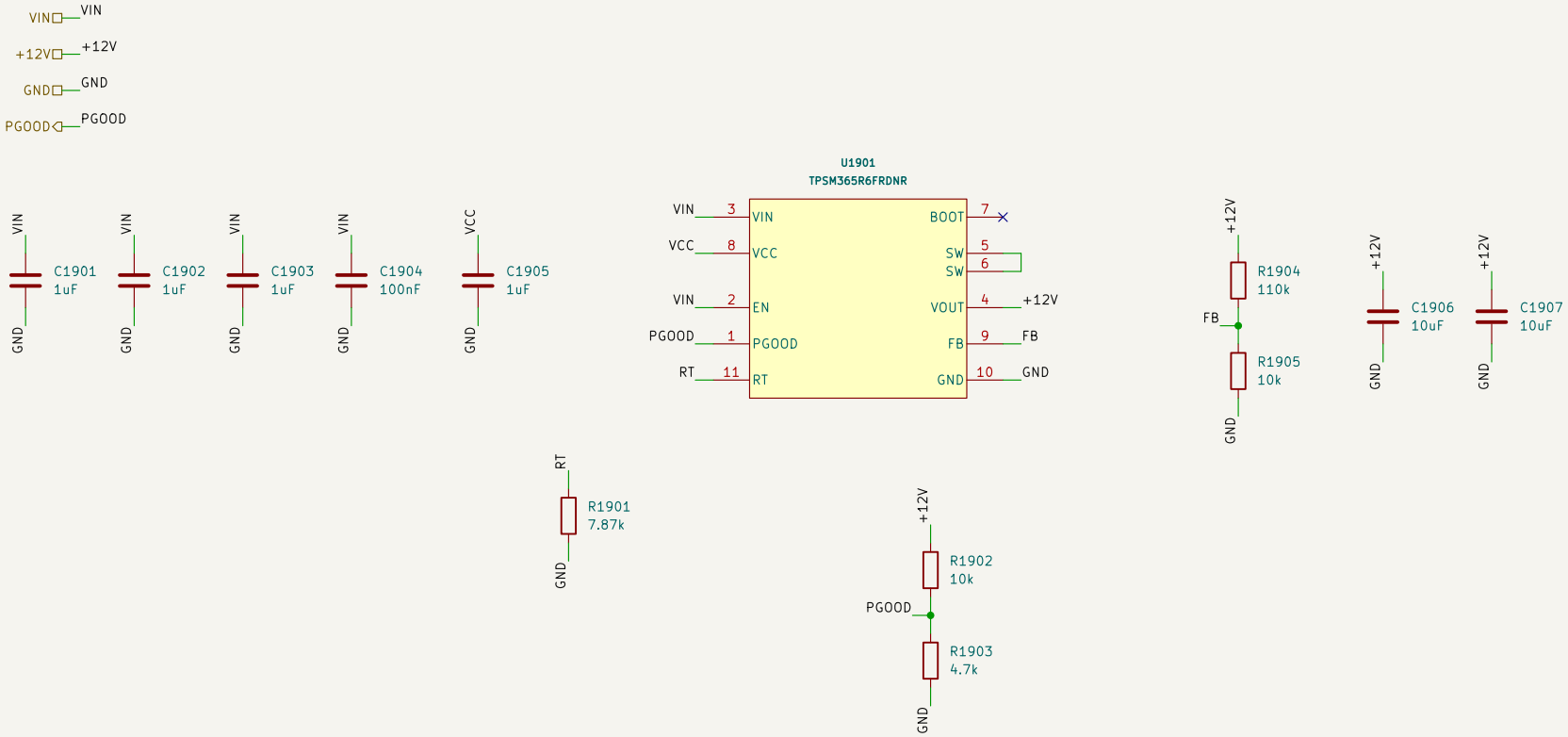






Janelia Research Campus		
Sheet: /brake_torque_sensor/		
File: brake_torque_sensor.kicad_sch		
Title: mouse-joystick-pcb		
Size: A4	Date: 2026-09-19	**Rev: 1.0**
KiCad E.D.A. 10.0.6		Id: 9/20


+36V
+12V
+5V
+3.3V
GND
SDA
SCL
BRAKE_ISENSE
BRAKE_FAULT
FB1801
600R@100MHz
FB1802
600R@100MHz
FB1803
600R@100MHz
PWR_FLAG
C1802 10uF
C1805 100nF
C1803 10uF
C1806 100nF
C1809 100nF
C1813 100nF
C1804 10uF
C1807 100nF
C1810 100nF
C1814 100nF
U1803A
SN74LVC2G17DRYR
U1803B
U1803C
C1817 470nF
C1818 100nF
D1802
D_Schottky_150V_1A
J1801
TBH-1x02-3.5mm
U1806
OPA197IDBVR
R1815 49.9k
R1811 49.9k
R1812 6.34k
C1816 100nF
C1820 10nF
U1804
OPA197IDBVR
R1817 10k
C1819 10pF
R1821 3.2k
U1801
DAC80501ZDGSR
C1811 0.47uF
R1807 1k
R1809 1k
R1813 1k
U1802
OPA197IDBVR
U1805
BUK7D36-60EX
R1814 49.9k
D1801
D_SCHOTTKY_20V_1A
R1816 10
R1818 110k
R1819 249
R1820 1k
D1803
TPD1E05U06DPYR
R1801 10k
R1803 3.2k
R1805 49.9k
R1804 3.2k
C1808 10nF
R1802 49.9k
R1806 49.9k
C1801 10nF
C1812 10nF
R1808 3.2k
R1810 3.2k
C1815 10nF
Copy or recreate old project circuits here, then save cleaned selections as design blocks.
Sheet: /brake_torque_sensor/brake_current_sink_0_49ma_dac80501/
File: brake_current_sink_0_49ma_dac80501.kicad_sch
Title: Orb Weaver Design Block Mining
Size: A4
Date: 2026-09-20
Rev:
KiCad E.D.A. 10.0.6
Id: 18/20



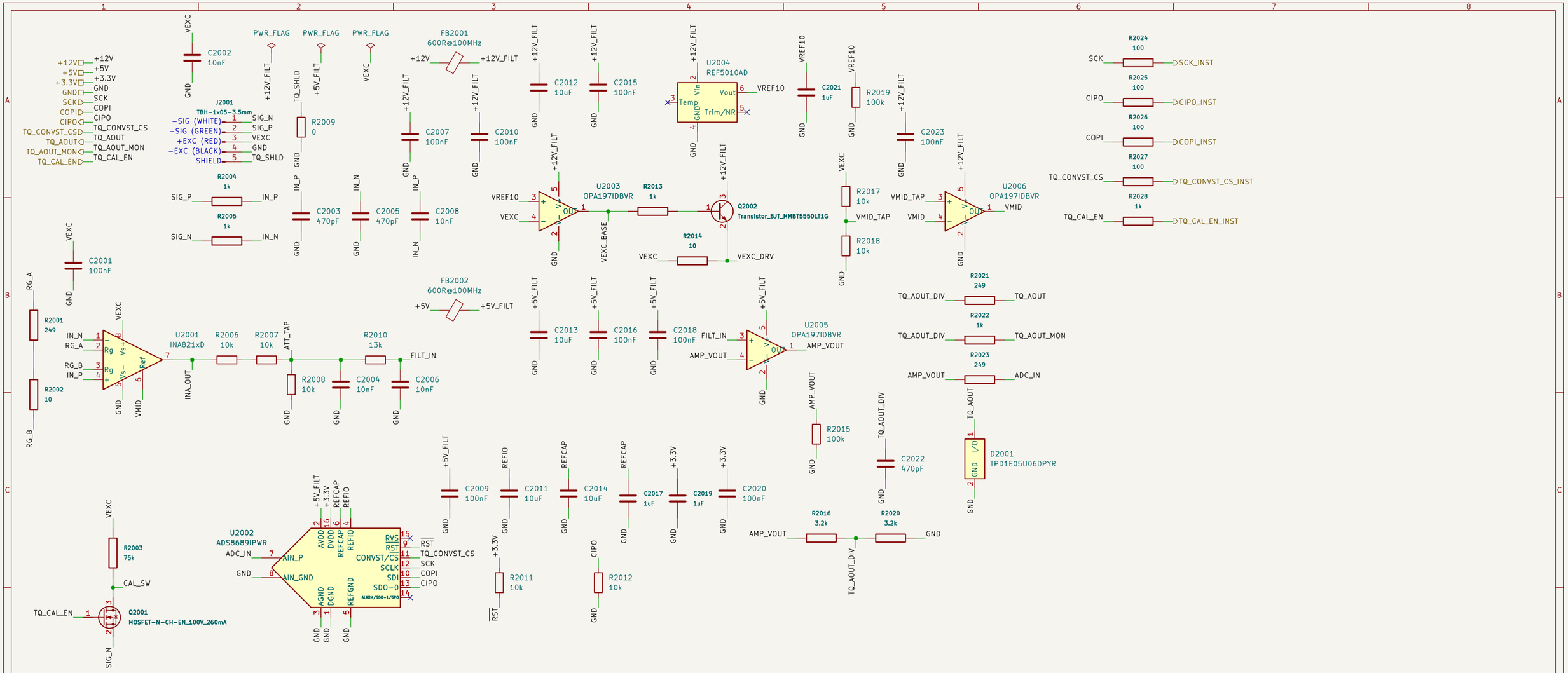
Copy or recreate old project circuits here, then save cleaned selections as design blocks.

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Sheet: /brake_torque_sensor/hv_to_12v_tpsm365r6/
File: hv_to_12v_tpsm365r6.kicad_sch

Title: mouse-joystick-pcb

Size: A4	Date: 2026-09-19	**Rev: 1.0**
KiCad E.D.A. 10.0.6		Id: 19/20

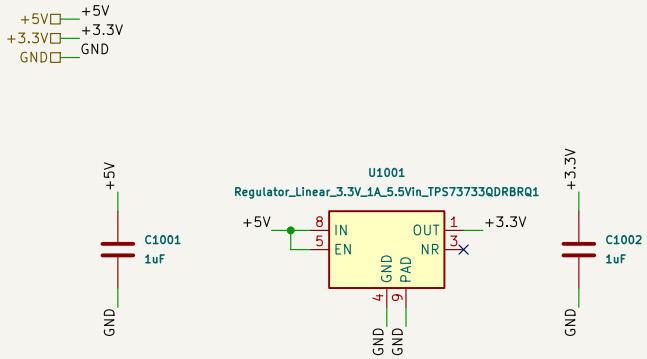


Copy or recreate old project circuits here, then save cleaned selections as design blocks.

Sheet: /brake_torque_sensor/torque_bridge_0_3v3_ina821/
File: torque_bridge_0_3v3_ina821.kicad_sch

Title: Orb Weaver Design Block Mining

Size: A3	Date: 2026-09-20	**Rev:**
KiCad E.D.A. 10.0.6		Id: 20/20



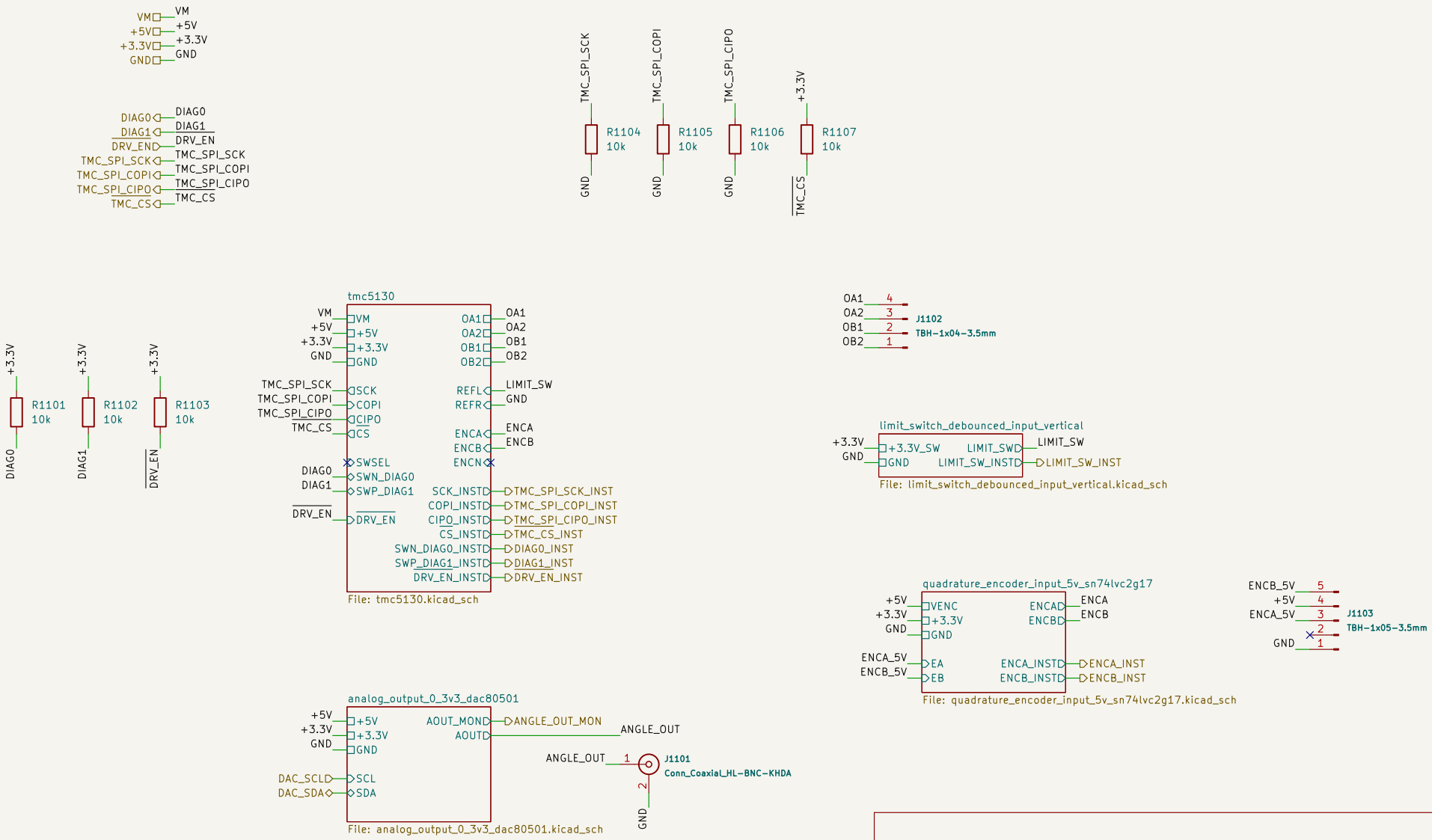
Copy or recreate old project circuits here, then save cleaned selections as design blocks.

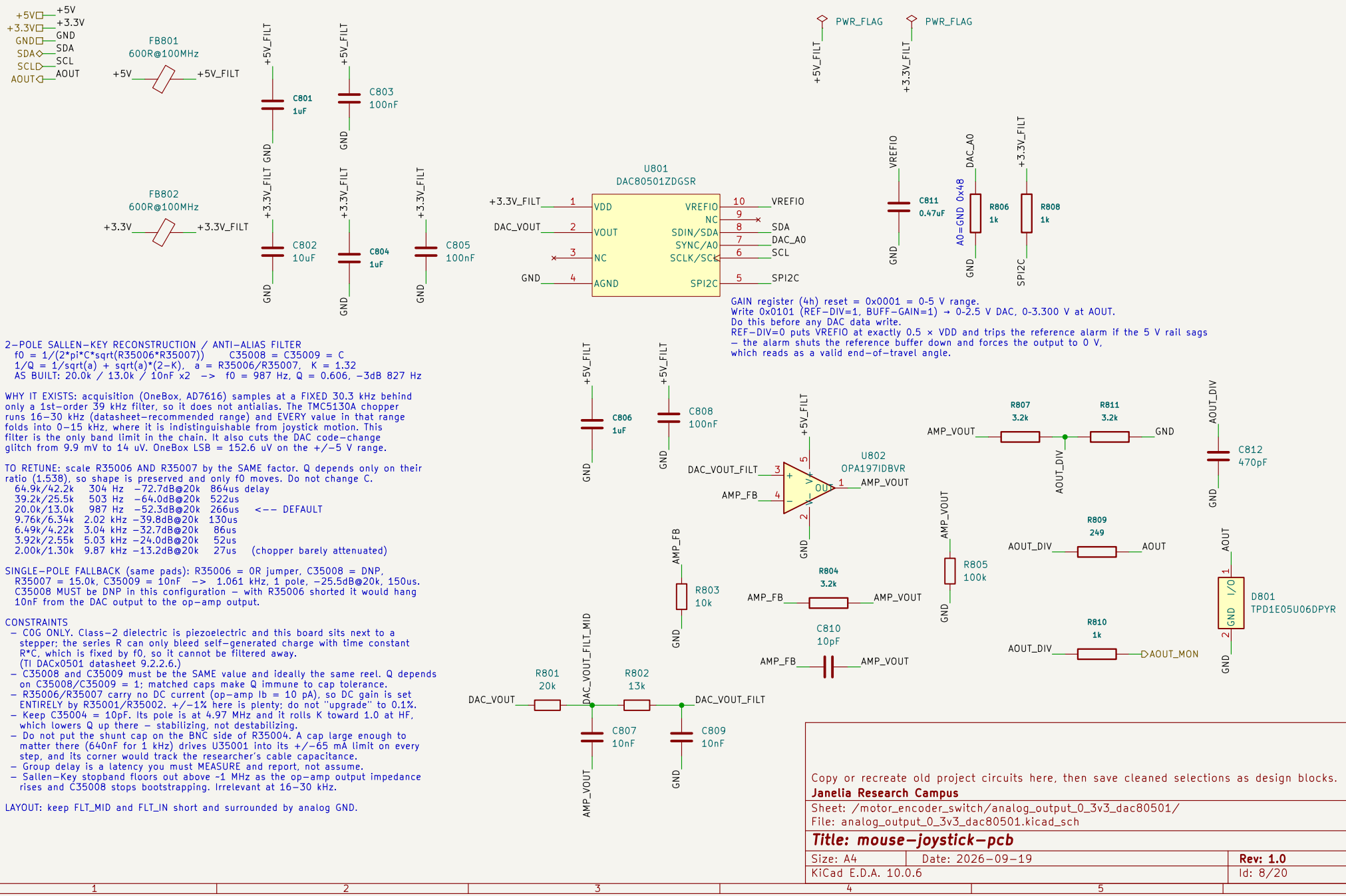
Janelia Research Campus

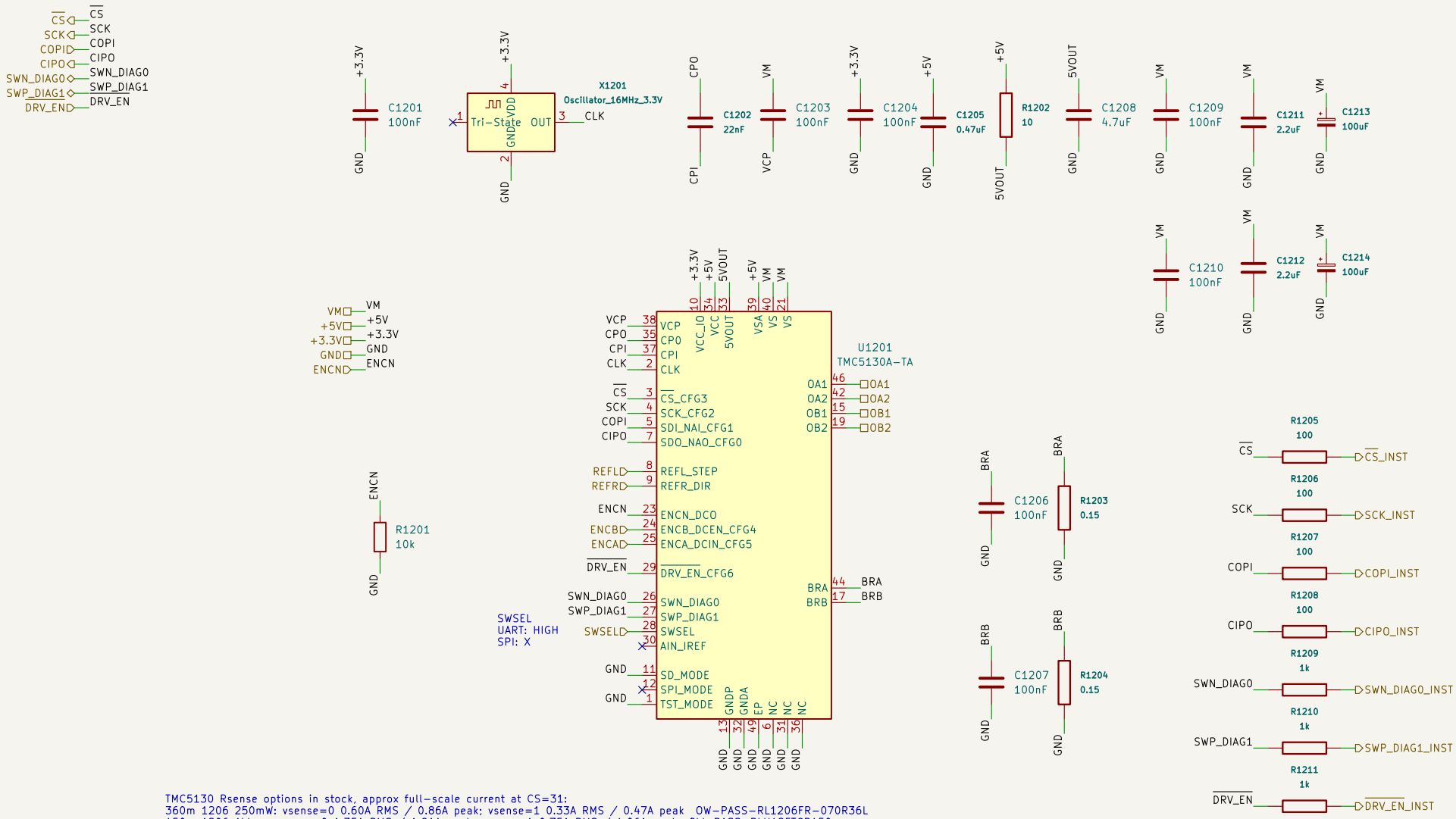
Sheet: /5v_to_3v3_tps73733_2/
File: 5v_to_3v3_tps73733_2.kicad_sch

Title: mouse-joystick-pcb

Size: A4	Date: 2026-09-19	**Rev: 1.0**
KiCad E.D.A. 10.0.6		Id: 10/20


VM
+5V
+3.3V
GND
DIAG0
DIAG1
DRV_EN
TMC_SPI_SCK
TMC_SPI_COPI
TMC_SPI_CIPO
TMC_CS
R1104 10k
R1105 10k
R1106 10k
R1107 10k
R1101 10k
R1102 10k
R1103 10k
tmc5130
SWSEL
SWN_DIAG0
SWP_DIAG1
SCK_INST
COPI_INST
CIPO_INST
CS_INST
SWN_DIAG0_INST
SWP_DIAG1_INST
DRV_EN_INST
OA1
OA2
OB1
OB2
REFL
REFR
ENCA
ENCB
ENCN
LIMIT_SW
File: tmc5130.kicad_sch
J1102
TBH-1x04-3.5mm
limit_switch_debounced_input_vertical
+3.3V_SW
LIMIT_SW
LIMIT_SW_INST
File: limit_switch_debounced_input_vertical.kicad_sch
quadrature_encoder_input_5v_sn74lvc2g17
VENC
+3.3V
ENCA_5V
ENCB_5V
EA
EB
ENCA_INST
ENCB_INST
File: quadrature_encoder_input_5v_sn74lvc2g17.kicad_sch
ENCB_5V
ENCA_5V
J1103
TBH-1x05-3.5mm
analog_output_0_3v3_dac80501
AOUT_MON
AOUT
ANGLE_OUT_MON
ANGLE_OUT
DAC_SCL
DAC_SDA
SCL
SDA
File: analog_output_0_3v3_dac80501.kicad_sch
J1101
Conn_Coaxial_HL-BNC-KHDA
Janelia Research Campus
Sheet: /motor_encoder_switch/
File: motor_encoder_switch.kicad_sch
Title: mouse-joystick-pcb
Size: A4
Date: 2026-09-19
Rev: 1.0
KiCad E.D.A. 10.0.6
Id: 11/20


+5V
+3.3V
GND
SDA
SCL
AOUT
FB801
600R@100MHz
FB802
600R@100MHz
C801
1uF
C802
10uF
C803
100nF
C804
1uF
C805
100nF
C806
1uF
C808
100nF
C807
10nF
C809
10nF
C810
10pF
C811
0.47uF
C812
470pF
U801
DAC80501ZDGSR
VDD
VOUT
NC
AGND
VREFIO
SDIN/SDA
SYNC/A0
SCLK/SCL
SPI2C
U802
OPA197IDBVR
R801
20k
R802
13k
R803
10k
R804
3.2k
R805
100k
R806
1k
R807
3.2k
R808
1k
R809
249
R810
1k
R811
3.2k
D801
TPD1E05U06DPYR
PWR_FLAG
PWR_FLAG
GAIN register (4h) reset = 0x0001 = 0-5 V range.
Write 0x0101 (REF-DIV=1, BUFF-GAIN=1) → 0-2.5 V DAC, 0-3.300 V at AOUT.
Do this before any DAC data write.
REF-DIV=0 puts VREFIO at exactly 0.5 × VDD and trips the reference alarm if the 5 V rail sags
– the alarm shuts the reference buffer down and forces the output to 0 V,
which reads as a valid end-of-travel angle.
2-POLE SALLEN-KEY RECONSTRUCTION / ANTI-ALIAS FILTER
f0 = 1/(2*pi*C*sqrt(R35006*R35007)) C35008 = C35009 = C
1/Q = 1/sqrt(a) + sqrt(a)*(2-K), a = R35006/R35007, K = 1.32
AS BUILT: 20.0k / 13.0k / 10nF x2 -> f0 = 987 Hz, Q = 0.606, -3dB 827 Hz
WHY IT EXISTS: acquisition (OneBox, AD7616) samples at a FIXED 30.3 kHz behind
only a 1st-order 39 kHz filter, so it does not antialias. The TMC5130A chopper
runs 16-30 kHz (datasheet-recommended range) and EVERY value in that range
folds into 0-15 kHz, where it is indistinguishable from joystick motion. This
filter is the only band limit in the chain. It also cuts the DAC code-change
glitch from 9.9 mV to 14 uV. OneBox LSB = 152.6 uV on the +/-5 V range.
TO RETUNE: scale R35006 AND R35007 by the SAME factor. Q depends only on their
ratio (1.538), so shape is preserved and only f0 moves. Do not change C.
64.9k/42.2k 304 Hz -72.7dB@20k 864us delay
39.2k/25.5k 503 Hz -64.0dB@20k 522us
20.0k/13.0k 987 Hz -52.3dB@20k 266us <-- DEFAULT
9.76k/6.34k 2.02 kHz -39.8dB@20k 130us
6.49k/4.22k 3.04 kHz -32.7dB@20k 86us
3.92k/2.55k 5.03 kHz -24.0dB@20k 52us
2.00k/1.30k 9.87 kHz -13.2dB@20k 27us (chopper barely attenuated)
SINGLE-POLE FALLBACK (same pads): R35006 = 0R jumper, C35008 = DNP,
R35007 = 15.0k, C35009 = 10nF -> 1.061 kHz, 1 pole, -25.5dB@20k, 150us.
C35008 MUST be DNP in this configuration – with R35006 shorted it would hang
10nF from the DAC output to the op-amp output.
CONSTRAINTS
– C0G ONLY. Class-2 dielectric is piezoelectric and this board sits next to a
stepper; the series R can only bleed self-generated charge with time constant
R*C, which is fixed by f0, so it cannot be filtered away.
(TI DACx0501 datasheet 9.2.2.6.)
– C35008 and C35009 must be the SAME value and ideally the same reel. Q depends
on C35008/C35009 = 1; matched caps make Q immune to cap tolerance.
– R35006/R35007 carry no DC current (op-amp Ib = 10 pA), so DC gain is set
ENTIRELY by R35001/R35002. +/-1% here is plenty; do not "upgrade" to 0.1%.
– Keep C35004 = 10pF. Its pole is at 4.97 MHz and it rolls K toward 1.0 at HF,
which lowers Q up there – stabilizing, not destabilizing.
– Do not put the shunt cap on the BNC side of R35004. A cap large enough to
matter there (640nF for 1 kHz) drives U35001 into its +/-65 mA limit on every
step, and its corner would track the researcher's cable capacitance.
– Group delay is a latency you must MEASURE and report, not assume.
– Sallen-Key stopband floors out above ~1 MHz as the op-amp output impedance
rises and C35008 stops bootstrapping. Irrelevant at 16-30 kHz.
LAYOUT: keep FLT_MID and FLT_IN short and surrounded by analog GND.
Copy or recreate old project circuits here, then save cleaned selections as design blocks.
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Sheet: /motor_encoder_switch/analog_output_0_3v3_dac80501/
File: analog_output_0_3v3_dac80501.kicad_sch
Title: mouse-joystick-pcb
Size: A4
Date: 2026-09-19
Rev: 1.0
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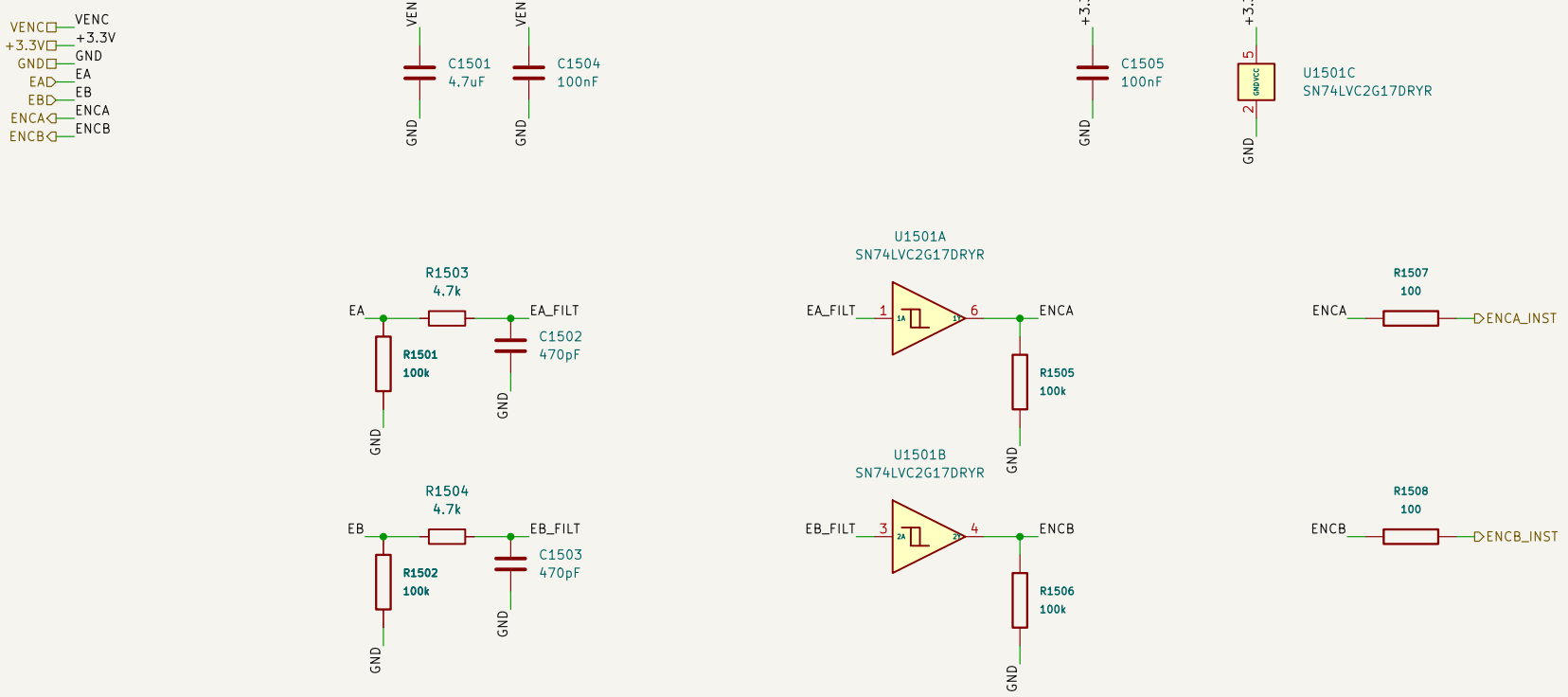
TMC5130 Rsense options in stock, approx full-scale current at CS=31:
360m 1206 250mW: vsense=0 0.60A RMS / 0.86A peak; vsense=1 0.33A RMS / 0.47A peak OW-PASS-RL1206FR-070R36L
150m 1206 1W: vsense=0 1.35A RMS / 1.91A peak; vsense=1 0.75A RMS / 1.06A peak OW-PASS-RLH12FTCR150
120m 1206 750mW: vsense=0 1.64A RMS / 2.32A peak; vsense=1 0.91A RMS / 1.29A peak OW-PASS-KRL1632E-M-R120-F-T5
Set lower current in firmware; choose Rsense so normal IRUN is ~16-31.
TMC5130 TQFP thermal behavior needs validation above ~1.2A RMS continuous.

Copy or recreate old project circuits here, then save cleaned selections as design blocks.

Sheet: /motor_encoder_switch/tmc5130/
File: tmc5130.kicad_sch

Title: Orb Weaver Design Block Mining

Size: A4	Date: 2026-09-19	Rev:
KiCad E.D.A. 10.0.6		Id: 12/20

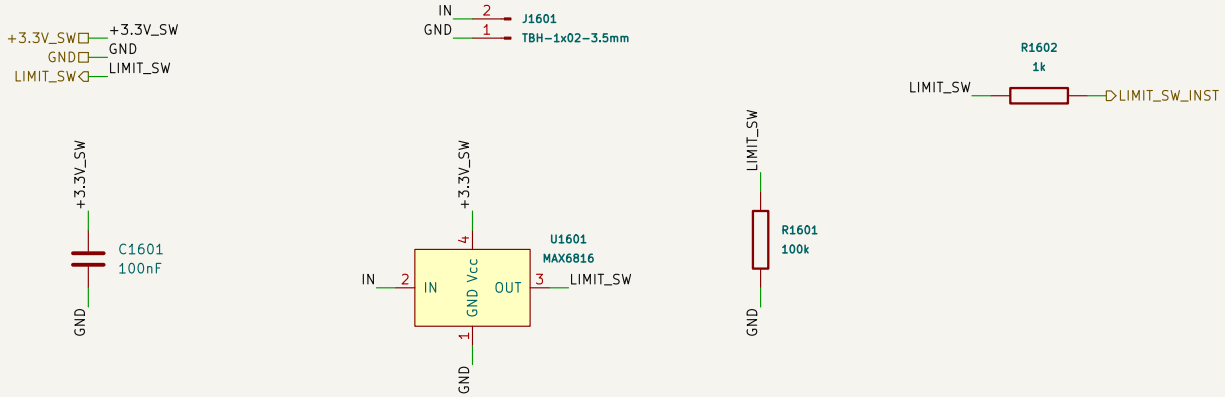


Copy or recreate old project circuits here, then save cleaned selections as design blocks.

Sheet: /motor_encoder_switch/quadrature_encoder_input_5v_sn74lvc2g17/
File: quadrature_encoder_input_5v_sn74lvc2g17.kicad_sch

Title: Orb Weaver Design Block Mining

Size: A4	Date: 2026-09-20	**Rev:**
KiCad E.D.A. 10.0.6		Id: 15/20

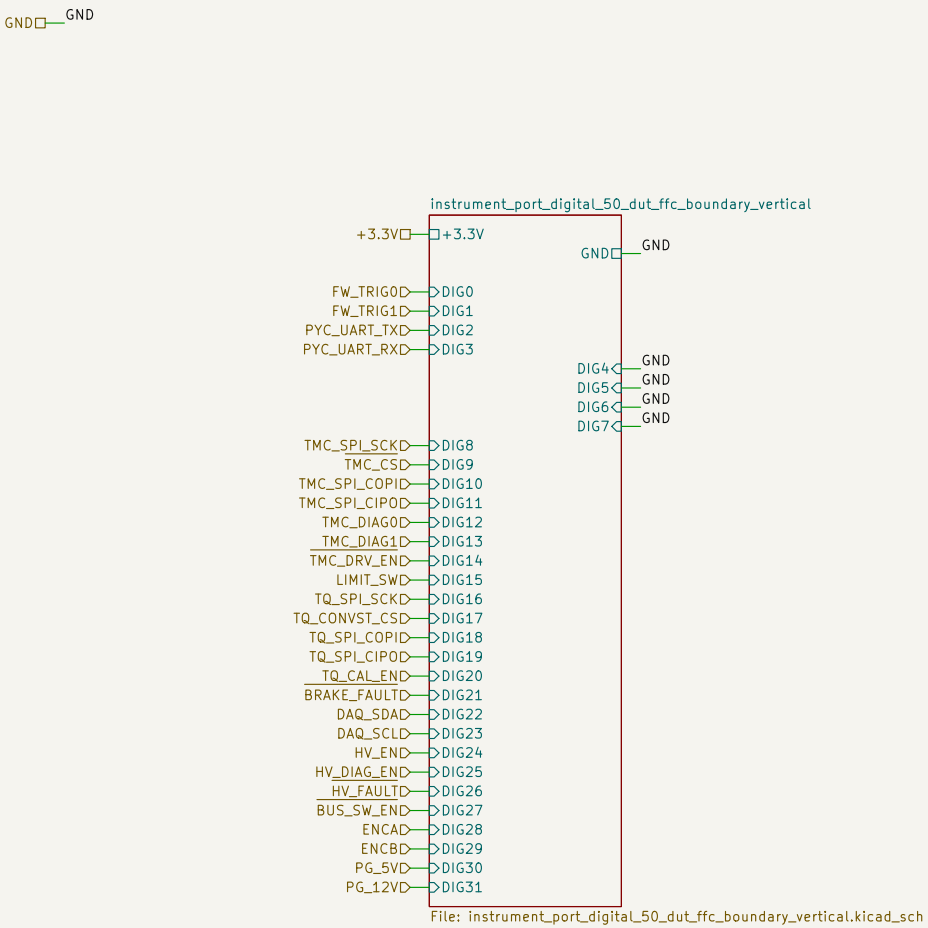


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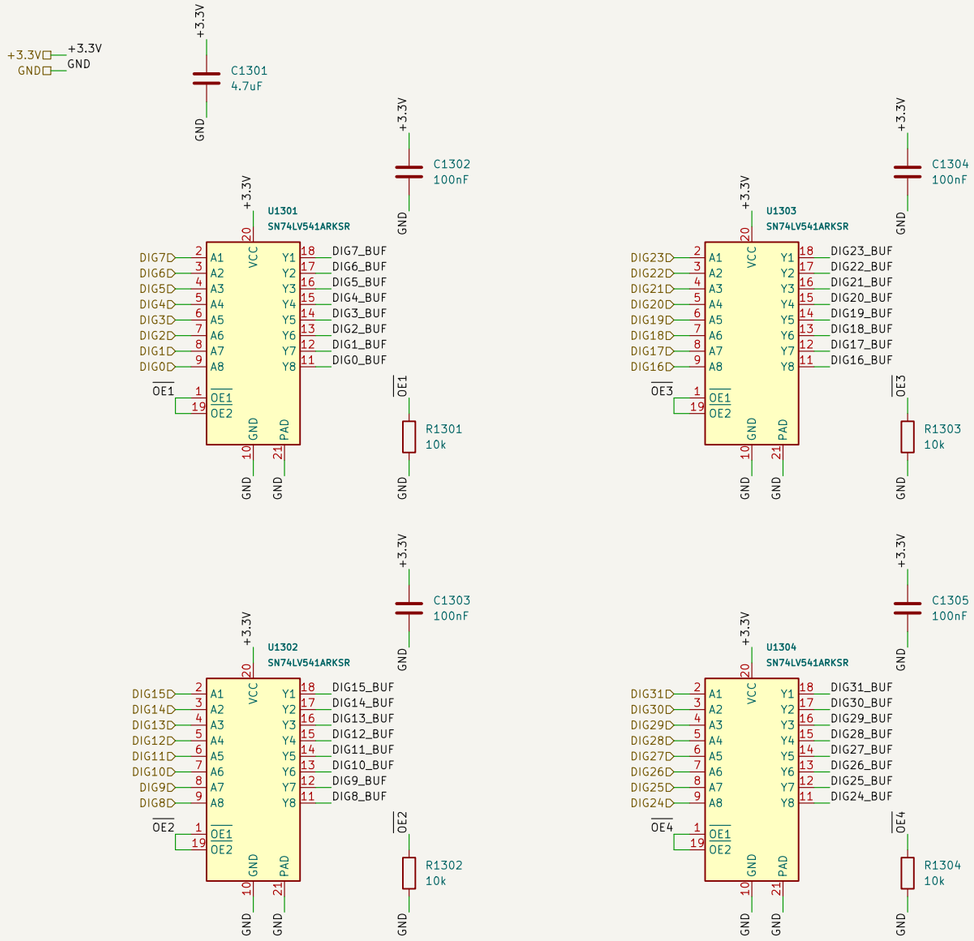
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File: limit_switch_debounced_input_vertical.kicad_sch

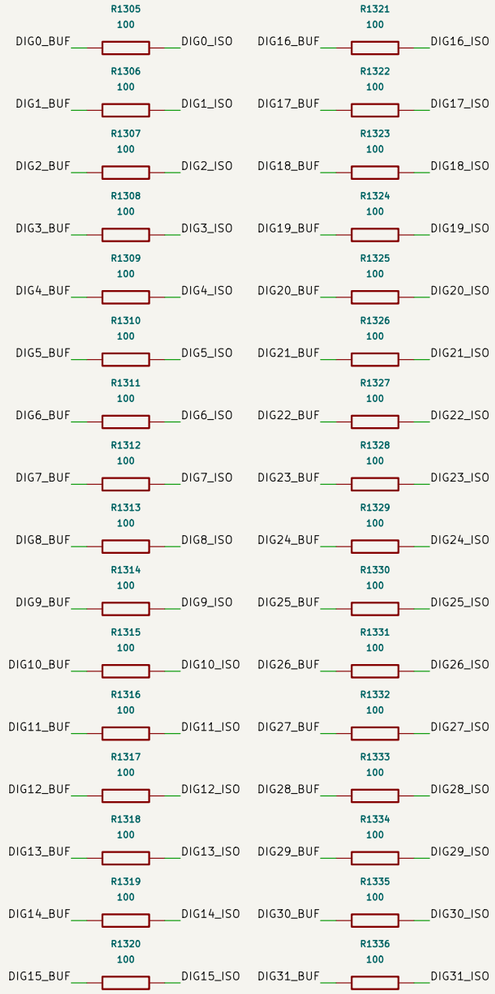
Title: Orb Weaver Design Block Mining

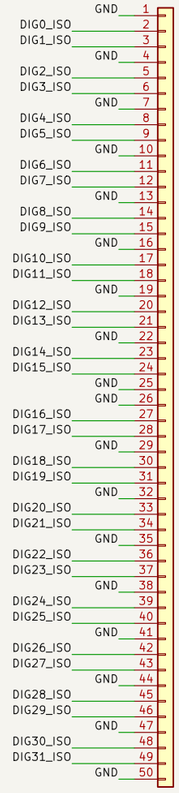
Size: A4	Date: 2026-09-20	**Rev:**
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Sheet: /instrument_port/ File: instrument_port.kicad_sch		
Title: mouse-joystick-pcb		
Size: A4	Date: 2026-09-19	**Rev: 1.0**
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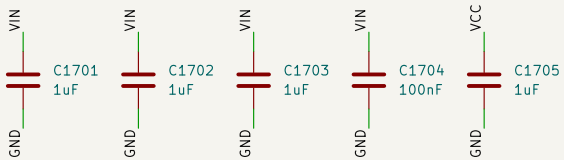
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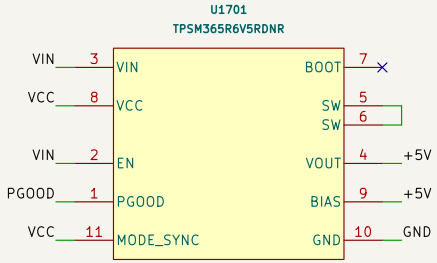
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File: instrument_port_digital_50_dut_ffc_boundary_vertical.kicad_sch

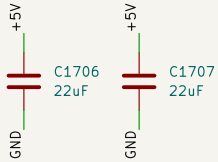
Title: mouse-joystick-pcb

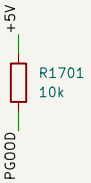
Size: A3	Date: 2026-09-19	**Rev: 1.0**
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Sheet: /hv_to_5v_tpsm365r6/
File: hv_to_5v_tpsm365r6.kicad_sch

Title: mouse-joystick-pcb

Size: A4	Date: 2026-09-19	**Rev: 1.0**
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